Reader Supplement to
Triangle Hypersurfaces and a Sharp Identifiability Boundary for Level-2 K3P Networks

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This supplement is a reader-facing map of the exact proof package. It gives the formulas, finite censuses, witness selections, and hash bindings needed to audit the article without reproducing raw JSON ledgers. All paths are relative to the project root `k3p_level2_identifiability_final/`. The machine-readable files, not line-broken values in this document, are canonical for exact replay.

1 Dependency graph

The necessity and sufficiency chains are:

```
K3P domain + root invariance
  -> true-cut rank <= 4 + generic noncut recovery
  -> Cut(target) subset Cut(source)
  -> crossing-split dichotomy + 204 pointwise one-active directions
  -> Cut(source) subset Cut(target)
  -> common labelled abstract bridge-incidence tree
  -> trivalent ordinary-versus-cycle decoration by the six-circuit separator
  -> three-sector bridge fibre + physical local product
  -> marginal submersion + finite target-type localization
  -> 14-orbit theorem + restoration + coherent probes
  -> necessity: topology equivalence modulo ordinary triangles

H14 quartic + three rank-14 orientation sections
  -> common relative triangle germ
  -> one labelled contextual contraction
  -> simultaneous principal/CT bridge gluing
  -> sufficiency and symmetric common full-dimensional germ

main classification + finite topology set + semialgebraic stratification
  -> proper exceptional set
  -> generic identifiability + terminating exact reconstruction

weak-not-strong topology census + Krawczyk equality box + two rank-15 maps
  -> base ambient-open common germ
  -> six-dimensional identical-cherry inverse
  -> all-n dimension 6n-3

all active theorem branches + machine-readable theorem specification
  -> integrated verification of K3P containment in the strongly tree-child
     level-2 class
```

```
-> 24/24 fail-closed integrated mutations
-> all mathematical checks pass; publication engineering remains separate
```

The cut-transfer chain uses neither a common bridge tree nor the fourteen-orbit classification. The local classification is invoked only after cut equality has been proved.

2 Theorem-to-certificate crosswalk

claim	minimal active evidence and replay boundary
K3P inverse Fourier formulas, $\mathcal{D}_{3,+}, \mathcal{D}_{3,\mathrm{CT}}$, subdivision, root movement	`model_domain/primary_exact_evidence.json`; primary exact replay.
Cycle–theta primitive topology, no-omnian grammar, and strong repairs	The article's cycle–theta primitive proposition gives the complete pole/source/sink case split. The hash-bound companion sources and `COMPANION_DEPENDENCY_LOCK.json` preserve provenance; model-specific K2P algebra is not active K3P evidence.
True-cut rank and generic noncut recovery	article derivation; frozen JC cut theorem only on the relatively open strict JC locus in the isotropic slice, not on an open subset of full K3P; `cut_recovery/adversarial/` preserves the failed universal shortcut.
204 pointwise one-active directions	`cut_recovery/strong_crossbridge/final_certificate/STRONG_CROSSBRIDGE_FINAL_CERTIFICATE.json` and its producer-independent K3P verifier. The active graph producer in `cut_recovery/strong_crossbridge/topology_regeneration/` derives the five templates, 72 labelled records, and literal switching signatures without reading the stored table; its fresh output is compared byte for byte with the K3P-bound input. The separate algebraic verifier then derives the 204 normalized directions and exact obstructions.
Strong-class cut-set equality under containment	`cut_recovery/strong_crossbridge/global_transfer/THEOREM_MANIFEST.json` and `cut_recovery/strong_crossbridge/global_transfer/GLOBAL_TRANSFER_AUDIT.md`; independent adversarial replay.
Trivalent ordinary-versus-cycle decoration	article proof after cut equality: each cubic tree–sunlet circuit is multiplied by a positive monomial under the three incidence gauges, so its zero/nonzero status is intrinsic; the strong theta grammar has at least four physical boundaries.
Tree–sunlet separation and triangle rank	`three_port/primary_exact_evidence.json`, `three_port/literal_separator_v2/K3P_TREE_SUNLET_LITERAL_SEPARATOR_V2.json`, its independent literal-map verifier, and its cyclic-renaming mutation suite. The frozen v1 file is retained only as superseded provenance.
H_{14}, irreducibility, smooth common germ, contextualization	`triangle_h14/K3P_H14_CONTEXT_CERTIFICATE.json`.
Three-sector bridge fibre and physical product	`bridge_fibre/K3P_BRIDGE_FIBRE_CERTIFICATE.json`.
Marginal descriptors and physical submersions	`marginals/K3P_MARGINAL_SUBMERSION_CERTIFICATE.json`.
Fourteen orbits and two sink swaps	The active full-universe replay under `four_port_atlas/full_universe_replay/` reconstructs all 405,216 presentations, all 27,834 post-topology cases, the exact rank and polynomial filters, the 2,540 restoration obligations, and the $40 = 38 + 2$ quotient. Its separately implemented verifier imports neither the producer nor its graph atlas. The representative exact certificates are bound by `four_port_atlas/primary_exact_evidence.json` and `clean_room/H21_01_TRANSPORT_AUDIT.json`.

claim	minimal active evidence and replay boundary
Fixed-full restoration	`restoration/RESTORATION_MANIFEST.json`, `restoration/K3P_RESTORATION_THEOREM_REPORT.md`, ledger, proof registry, `restoration/K3P_RESTORATION_INDEPENDENT_VERIFICATION.json`, and `restoration/K3P_RESTORATION_MUTATION_CERTIFICATE.json`; the independent replay passes and all 20 fail-closed restoration mutations are rejected.
One-/two-port coherence	`probes/K3P_PROBE_COHERENCE_CERTIFICATE.json`, the separately implemented all-row semantic replay `probes/verify_k3p_probes_semantic.py`, its report `probes/K3P_PROBE_SEMANTIC_VERIFICATION.json`, the coherently resealed mutation certificate `probes/K3P_PROBE_SEMANTIC_MUTATIONS.json`, the legacy streaming replay, and the v2 K3P restoration binding `probes/RESTORATION_MANIFEST.json`.
Global gluing, genericity, reconstruction	`global_infrastructure/K3P_GLOBAL_GLUE_AND_RECONSTRUCTION_CERTIFICATE.json`, `global_infrastructure/GLOBAL_INFRASTRUCTURE_MANIFEST.json`, and its independent replay, read together with the active cut-transfer and restoration bindings. The integrated primary gate passes all 28 entries.
K3P containment classification in the strongly tree-child level-2 class	The integrated classification-verifier report named in `ACTIVE_MANIFEST.json` verifies the principal-domain equivalence in the article's main theorem and its strict continuous-time corollary; the corresponding mutation report rejects all 24 integrated fail-closed mutations.
Weak-class sharpness and all n	`sharpness/K3P_SHARPNESS_KRAWCZYK_CERTIFICATE.json` and `sharpness/K3P_SHARPNESS_TOPOLOGY_ALL_N_CERTIFICATE.json`.
Outer tree–double-theta obstruction, including strict-CT deformation	`input_frozen/referenced_chat_manuscripts/tree_theta_collision_source.tex` contains the exact tangent and analytic implicit-function argument; `topology/primary_double_theta_evidence.json` and `topology/primary_rooting_census_evidence.json` bind the collision rank, dimension, and exclusion from weak tree-childness in the active primary replay.

3 Exact model domains

For an edge spectrum $(1, c, g, t)$, inverse Fourier transformation gives

$$\frac{1}{4}(1+c+g+t), \quad \frac{1}{4}(1+c-g-t), \quad \frac{1}{4}(1-c+g-t), \quad \frac{1}{4}(1-c-g+t).$$

Thus

$$\mathcal{D}_{3,+} = \{(c,g,t) \in (0,1)^3 : 1+c-g-t > 0,\ 1-c+g-t > 0,\ 1-c-g+t > 0\}.$$

Strict continuous time is

$$\mathcal{D}_{3,\mathrm{CT}} = \{(c,g,t) \in (0,1)^3 : c > gt,\ g > ct,\ t > cg\}.$$

For strict subdivision, choose

$$r > \max\{c, g, t, g+t-c, c+t-g, c+g-t\}$$

in the principal domain, or additionally $r > \max\{gt/c, ct/g, cg/t\}$ in continuous time, and use

$$(c,g,t)=(r,r,r)\odot(c/r,g/r,t/r).$$

For a serial class the selected descriptor is

$$((c_i,g_i,t_i))_{i=1}^m\mapsto(\prod_i c_i,\prod_i g_i,\prod_i t_i).$$

Its Jacobian has three disjoint positive rows and rank three.

4 Balanced noncut compression handoff

For the reverse cut inclusion, a balanced two-color assignment on one active cycle or theta factor is reduced without changing its status. Retain the primitive core, a rigid support Q consisting of one complete minimum strong repair and every path-sink child, and at least two actual labels of each color. Since $|Q|\le 4$, choosing at most four further actual labels gives at most eight ports; all other compulsory roles remain as zero-character completion ports. No blob edge is a bridge and no pendant arm isolates a complete color class, so the restriction remains noncut.

If a segment has three color runs c,d,c, the middle attachment lies on the path between the outer two in every switching, giving a direct all-switching obstruction. Otherwise, in the already bounded restriction, retain one actual label from each selected run, with palette

$$(),\quad (0),\quad (1),\quad (0,1),\quad (1,0),$$

and double a singleton run when required to retain two labels of a color. Any coloring displayed by every switching before restriction would survive this reduction. Independent graph/switching enumeration finds zero survivors in the compressed palette. A failing switching and the ordinary tree quartet criterion therefore produce two actual labels of each color in one of the finite wrong-quartet directions.

The active census checksum is five primitive cores, 72 active-labelled records, 216 split entries, 12 displayed-by-all entries, and 204 certified wrong directions. These counts verify the exhaustive replay; the combinatorial reduction above is the premise of the handoff.

5 The H_{14} coordinate chart and quartic

The conservation-supported coordinate order is

$$\begin{aligned}&(000,0CC,0GG,0TT,C0C,CC0,CGT,CTG,\\&\ G0G,GCT,GG0,GTC,T0T,TCG,TGC,TT0).\end{aligned}$$

With $q_{000}=1$, the ambient chart is $\mathbb{A}^{15}$. The exact quartic is

$$\begin{aligned}F_{H_{14}}=\;&q_{000}q_{CGT}q_{GTC}q_{TCG}-q_{000}q_{CTG}q_{GCT}q_{TGC}\\&-q_{0CC}q_{CGT}q_{G0G}q_{TT0}+q_{0CC}q_{CTG}q_{GG0}q_{T0T}\\&+q_{0GG}q_{C0C}q_{GCT}q_{TT0}-q_{0GG}q_{CC0}q_{GTC}q_{T0T}\\&-q_{0TT}q_{C0C}q_{GG0}q_{TCG}+q_{0TT}q_{CC0}q_{G0G}q_{TGC}.\end{aligned}$$

The common tensor in this order is

$$\left(1,\tfrac{1}{12},\tfrac{1}{12},\tfrac{1}{12},\tfrac{1}{12},\tfrac{1}{12},\tfrac{1}{48},\tfrac{1}{48},\tfrac{1}{12},\tfrac{1}{48},\tfrac{1}{12},\tfrac{1}{48},\tfrac{1}{12},\tfrac{1}{48},\tfrac{1}{48},\tfrac{1}{12}\right).$$

At this point $F=0$ and six normalized gradient coordinates are nonzero, each with absolute value $1/6912$. The irreducibility certificate treats the quartic as linear in q_{0CC}, with coefficient $-q_{CGT}q_{G0G}q_{TT0}+q_{CTG}q_{GG0}q_{T0T}$; a specialization kills that coefficient but leaves remainder -1. One hand-checkable choice is $q_{000}=q_{CTG}=q_{GCT}=q_{TGC}=1$, with every other coefficient-ring variable in the quartic set to zero.

6 Triangle rank witnesses

All three common preimages have edge order (a, b, c, d, e, f), spectra $(1/2, 1/2, 1/2)$ on the first five edges, $(1/3, 1/3, 1/3)$ on f, and inheritance $1/2$.

orientation	normalized output rows	parameter columns	determinant
1	$(1, 2, 3, 4, 6, 8, 7, 10, 12, 9, 13, 5, 11, 15)$	$(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 16, 17)$	$-1/760840571584512$
2	$(4, 8, 12, 5, 7, 3, 6, 1, 10, 9, 11, 2, 13, 15)$	same	$-1/760840571584512$
3	$(4, 8, 12, 5, 6, 3, 7, 1, 10, 11, 9, 2, 13, 15)$	same	$1/760840571584512$

The tree rank-9 witness uses output rows $(4, 8, 12, 5, 6, 3, 7, 1, 10)$, parameter columns $0, \ldots, 8$, and determinant $8192/28940625$.

7 Tree–sunlet six-circuit certificate

Write $I(L; R) = \prod_{u \in L} q_u - \prod_{v \in R} q_v$ and use the literal map

$$q_{xyz} = a_x b_y c_z [\lambda f_y d_z + (1 - \lambda) f_x e_z].$$

Put $\mu = \lambda(1 - \lambda)$ and

$$M_1 = a_C a_G b_G b_T c_C c_T, \quad M_2 = a_C a_T b_G b_T c_C c_G, \quad M_3 = a_G a_T b_C b_G c_C c_T,$$
$$M_4 = a_G a_T b_C b_T c_C c_G, \quad M_5 = a_C a_G b_C b_T c_G c_T, \quad M_6 = a_C a_T b_C b_G c_G c_T.$$

The six exact left-minus-right pullbacks are:

	$L; R$	exact literal pullback
I_1	$000, CGT, GTC; 0TT, C0C, GG0$	$\mu M_1(f_C f_T - f_G)(d_C e_T - d_T e_C f_G)$
I_2	$000, CTG, TGC; 0GG, C0C, TT0$	$\mu M_2(f_C f_G - f_T)(d_C e_G - d_G e_C f_T)$
I_3	$000, GCT, TGC; 0CC, GG0, T0T$	$-\mu M_3(f_C f_T - f_G)(d_C e_T f_G - d_T e_C)$
I_4	$000, GTC, TCG; 0CC, G0G, TT0$	$-\mu M_4(f_C f_G - f_T)(d_C e_G f_T - d_G e_C)$
I_5	$000, CTG, GCT; 0TT, CC0, G0G$	$-\mu M_5(f_C - f_G f_T)(d_G e_T - d_T e_G f_C)$
I_6	$000, CGT, TCG; 0GG, CC0, T0T$	$\mu M_6(f_C - f_G f_T)(d_G e_T f_C - d_T e_G)$

All six vanish on a tree. On a strict sunlet all μM_i are positive. If the shared first factor in a pair is nonzero, simultaneous vanishing of (I_1, I_3), (I_2, I_4), or (I_5, I_6) forces, respectively, $f_G^2 = 1$, $f_T^2 = 1$, or $f_C^2 = 1$, all impossible. If all three composition margins vanish, then $f_G = f_C f_T$, $f_T = f_C f_G$, and $f_C = f_G f_T$, so $p = f_C f_G f_T$ satisfies $p = p^2$ with $p \in (0, 1)$, again impossible. Hence $\sum_{j=1}^{6} I_j^2 > 0$. The active v2 certificate expands these six eight-term identities directly from the displayed map and rejects any unannounced cyclic renaming of d, e, f.

8 Three-sector incidence normalizers

Call a retained component marked if it carries a retained physical leaf boundary block, and unmarked otherwise. In each nonzero sector the physical block supplies the compensating character for a one-character anchor at each incident bridge. Every unmarked retained strong-class component has bridge degree at least three; the only possible degree-two theta stabilizer is the two-boundary $K_4 - e$ factor, which has no tree-child rooting.

For an unmarked component of degree $d \geq 3$, the one-sector exponent matrix has rows $e_1 + e_2, e_1 + e_3, e_2 + e_3, e_1 + e_4, \ldots, e_1 + e_d$. Its leading block is

$$B_3 = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad \det B_3 = -2,$$

and every later row introduces one new column, so $\operatorname{rank} B_d = d$. The full K3P normalizer matrix is

$$\operatorname{diag}(B_d, B_d, B_d),$$

with rank $3d$ and leading determinant -8. In each sector,

$$a_1 = \sqrt{r_{12}r_{13}/r_{23}}, \quad a_2 = \sqrt{r_{12}r_{23}/r_{13}}, \quad a_3 = \sqrt{r_{13}r_{23}/r_{12}}, \quad a_k = r_{1k}/a_1.$$

9 Capped simultaneous bridge gluing

On finitely many compactly contained local germs, let $0 < L \le A_h \le U$ bound every endpoint-incidence product. The active construction uses

$$\varepsilon = \min\left\{\frac{1}{4}, \frac{L^2}{8U}\right\}, \qquad z = (\varepsilon, \varepsilon, \varepsilon), \qquad x_h = \frac{\varepsilon}{A_h}.$$

The effective coordinate z has principal margins at least $3/4$ and continuous-time margins at least $3\varepsilon/4$. The actual physical bridge satisfies

$$\frac{\varepsilon}{U} \le x_h \le \frac{\varepsilon}{L} \le \frac{1}{8},$$

so its principal margins are at least $3/4$, while cyclically

$$x_C - x_G x_T \ge \frac{\varepsilon}{U} - \frac{\varepsilon^2}{L^2} \ge \frac{7\varepsilon}{8U} > 0.$$

Strict uniformity leaves a common three-dimensional neighborhood in which the effective coordinates vary freely and $x_h = z_h/A_h$ remains physical; this supplies the bridge-coordinate rank used in the global germ.

10 Genericity incidence argument

Let

$$R_N = \{\theta : \operatorname{rank} D\Phi_N(\theta) < d_N\}.$$

A finite Nash constant-rank stratification of R_N gives $\dim \Phi_N(R_N) \le d_N - 1$. Thus a full d_N-dimensional physical intersection cannot lie entirely in the source rank-drop image. For a competitor N', stratify the incidence correspondence

$$Z_{N'} = \{(q, \theta') : q = \Phi_{N'}(\theta'),\ q \in \mathcal{M}_{3,+}(N)\}$$

compatibly with the regular source image and target rank loci. If the physical intersection had dimension d_N, one incidence stratum would project with rank d_N onto a relatively open regular source-image germ. The analytic constant-rank theorem gives a physical target section $s(q)$; on a regular source-parameter neighborhood, $\sigma = s \circ \Phi_N$ is exactly the section required by directed containment.

Semialgebraic dimension agrees with real-Zariski-closure dimension. If $A = \mathbb{R}[x]/I_{\mathbb{R}}(C)$, the finite faithfully flat integral extension $A \to A \otimes_{\mathbb{R}} \mathbb{C}$ preserves Krull dimension, so the corresponding complex closure is also proper. Explicitly, E_N is the union inside $\mathcal{V}_N$ of the complex closure of the rank-drop image, the complex closures of all inequivalent physical intersections, the singular locus of $\mathcal{V}_N$, and every certified reconstruction-test hypersurface. These proper closed sets form a proper exceptional set rather than merely a measure-zero set.

11 Fourteen-orbit and 38+2 census

The symbols S0–S4, T80, and T822 are deterministic compiler enumeration keys, not additional topology types. Explicitly, S0–S4 denote (θ_0, r_0), (θ_0, r_1), (θ_1, r_0), (θ_1, r_1), and (θ_3, r_0); T80 denotes the (θ_0, r_1) completion, and T822 the (θ_3, r_1) completion with selected incoming port, both sink-child ports, and the segment-4 repair port. A four-digit word gives the literal port relabeling $i \mapsto p_i$.

orbit	source	target; port permutation	rank	raw	result
H21-01	S1 (θ_0, r_1)	T80 $(\theta_0, r_1; \texttt{0132})$	$21 \to 21$	4	quartic
H21-02	S1	T80 `0213`	$21 \to 21$	4	rank
H21-03	S1	T80 `0231`	$21 \to 21$	4	quartic
H21-04	S1	T80 `0312`	$21 \to 21$	4	quartic
H21-05	S1	T80 `0321`	$21 \to 21$	2	quartic
H21-06	S1	T80 `1032`	$21 \to 21$	4	quartic
L20-01	S0 (θ_0, r_0)	T822 $(\theta_3, r_1; \texttt{1203})$	$20 \to 24$	2	quartic
L20-02	S0	T822 `1230`	$20 \to 24$	2	rank
L21a-01	S2 (θ_1, r_0)	T822 `0123`	$21 \to 24$	2	quartic
L21a-02	S2	T822 `0132`	$21 \to 24$	2	rank
L21b-01	S3 (θ_1, r_1)	T822 `0123`	$21 \to 24$	2	quartic
L21b-02	S3	T822 `0132`	$21 \to 24$	2	rank
L23-01	S4 (θ_3, r_0)	T822 `0123`	$23 \to 24$	2	rank
L23-02	S4	T822 `0132`	$23 \to 24$	2	quartic

Raw membership sums to 38. Two rank-24 (θ_3, r_1) sink swaps on the T822 rigid support, with port permutations `0132` and `1023`, are separately quartic-separated. Thus $40 = 38 + 2$, with zero unresolved rows.

12 Two sink-swap records

The two final post-filter records not absorbed into the fourteen canonical orbits compare two labelled presentations of the same rigid (θ_3, r_1) support just described. Their port permutations are `0132` and `1023`. Neither transport is a labelled mixed-graph isomorphism. For each record the target pullback of Q_{23} is zero, while the source pullback has 160 terms and is nonzero at a strict rational point of minimum physical margin 7/31. The common source-pullback SHA-256 is abbreviated as `c1f4289be568...0c6f4a`. Consequently neither record yields a new symmetric move or a directed containment.

13 Polynomial separator data

For four-port coordinates, indices $0, \dots, 63$ use the lexicographic conservation-supported order

$$
\begin{multlined}
0000, 00CC, 00GG, 00TT, 0C0C, 0CC0, 0CGT, 0CTG, 0G0G, 0GCT, 0GG0, 0GTC, \\
0T0T, 0TCG, 0TGC, 0TT0, C00C, C0C0, C0GT, C0TG, CC00, CCCC, CCGG, CCTT, \\
CG0T, CGCG, CGGC, CGT0, CT0G, CTCT, CTG0, CTTC, G00G, G0CT, G0G0, G0TC, \\
GC0T, GCCG, GCGC, GCT0, GG00, GGCC, GGGG, GGTT, GT0C, GTC0, GTGT, GTTG, \\
T00T, T0CG, T0GC, T0T0, TC0G, TCCT, TCG0, TCTC, TG0C, TGC0, TGGT, TGTG, \\
TT00, TTCC, TTGG, TTTT.
\end{multlined}
$$

Each body below is a signed sum of eight degree-four monomials. A tuple (i, j, k, l) denotes $q_i q_j q_k q_l$.

body	signed index monomials
$Q_{H,a}$	$+(0,24,44,52) - (0,28,36,56) - (4,24,32,60) + (4,28,40,48)$
	$+(8,16,36,60) - (8,20,44,48) - (12,16,40,52) + (12,20,32,56)$
$Q_{H,b}$	$+(0,18,35,49) - (0,19,33,50) - (1,18,32,51) + (1,19,34,48)$
	$+(2,16,33,51) - (2,17,35,48) - (3,16,34,49) + (3,17,32,50)$
Q_{20}	$+(0,10,35,61) - (0,11,34,61) + (0,14,41,51) - (0,15,41,50)$
	$-(1,10,35,60) + (1,11,34,60) - (1,14,40,51) + (1,15,40,50)$
Q_{21}	$+(0,10,44,55) + (0,11,38,60) - (0,14,40,55) - (0,15,38,56)$
	$-(4,10,44,51) - (4,11,34,60) + (4,14,40,51) + (4,15,34,56)$

body	signed index monomials
Q_{23}	$+(0,8,45,53)+(0,9,37,60)-(0,12,37,57)-(0,13,40,53)$ $-(5,8,45,48)-(5,9,32,60)+(5,12,32,57)+(5,13,40,48)$

The per-orbit transport and nonvanishing summary is:

orbit	body	marginal/representative transport	source terms	strict margin	source pullback SHA-256
H21-01	$Q_{H,a}$	omit 2; retain $(0,1,3)$	1080	2/9	`ef5ed407624b...f451f81f`
H21-03	$Q_{H,b}$	omit 1; retain $(0,2,3)$	192	2/9	`14f7d8b5683d...a8b9da28`
H21-04	$Q_{H,a}$	omit 2; retain $(0,1,3)$	1080	2/9	`ef5ed407624b...f451f81f`
H21-05	$Q_{H,b}$	omit 1; retain $(0,2,3)$	192	2/9	`14f7d8b5683d...a8b9da28`
H21-06	$Q_{H,a}$	omit 2; retain $(0,1,3)$	1080	2/9	`ef5ed407624b...f451f81f`
L20-01	Q_{20}	representative 1203	270	5/23	`34aee883f27a...502a371`
L21a-01	Q_{21}	representative 0123	270	7/31	`ff97c8b01748...c97521d0`
L21b-01	Q_{21}	representative 0123	270	5/23	`ff97c8b01748...c97521d0`
L23-02	Q_{23}	representative 0132	96	2/9	`fd18a85829e2...67bbe338`

All H21 character transports are the identity on the ordered labels C, G, T; only physical ports move. Every target pullback is exactly zero. The two sink swaps use Q_{23}, have 160-term source pullbacks, strict margin 7/31, and common pullback hash `c1f4289be568...0c6f4a`. Full hashes and exact rational source values are in the bound JSON files; abbreviations above are for page fit only.

14 Directed-rank witnesses

The table gives the selected output-row set, source minor columns, strict source rank, target generator upper bound, and target compression. Exact rational determinants and labels are in `input_frozen/k3p_cloud_artifacts/k3p_directed_rank_obstructions.json` (SHA-256 `fa0ac74cde903edc422a90b5e490bd639b2e2b4c758d9d3ec10a794f1a044f42`).

orbit	selected output rows	source columns	gap	target factorization
H21-02	$(3,12,15,20,27,39,$ $40,48,51,60,63)$	$(2,3,4,5,6,7,$ $8,11,14,20,23)$	$11>10$	ten generators $U, V, Z, D, I, A_0, B_0, A, B, \rho$, saturated at $e_{2C}e_{2G}DI \neq 0$
L20-02	$(5,10,15,17,20,27,30,34,$ $39,40,45,51,54,57,60)$	$(3,4,5,6,7,8,9,$ $10,11,12,13,14,18,19)$	$14>12$	ordinary-sunlet compression; omit port 3
L21a-02	$(5,15,17,20,27,39,$ $40,45,51,57,60)$	$(3,4,5,6,7,8,$ $9,10,11,15,16)$	$11>10$	sunlet generators with A_G, B_G absent; omit port 3
L21b-02	same as L21a-02	same	$11>10$	same ten-generator support; omit port 3
L23-01	$(4,8,12,16,20,24,28,32,$ $36,40,44,48,52,56,60)$	$(3,4,5,6,7,8,9,$ $10,11,12,13,14,21,22)$	$14>12$	ordinary-sunlet compression; omit port 2

The exact target minors have ranks $10, 12, 10, 10, 12$, respectively, and the function-field formulas prove these are upper bounds, not merely sampled ranks. These are directional selected-projection obstructions, not total model-dimension comparisons: a target can have equal or larger total rank while its indicated observable projection has function-field dimension at most 10 or 12. The H21 function-field calculation is made after saturation away from the declared physical factors, which are nonzero throughout the strict physical domain.

For H21-02, L21a-02, and L21b-02, the source minor uses the complete displayed row set. For L20-02 and L23-01, it uses the displayed set with row 60 deleted. The bound JSON records the determinant-sensitive row order and the exact nonzero rational determinant for every source minor.

15 Restoration census

certificate	minimal first layer	complete redundant forest
displayed quartet	35,758	36,006
K3P tree–sunlet SOS	606	614
K3P multihomogeneous quadratic	148	148
active K3P marginal quartic	56	56
edges/terminal rows	36,568	36,824

The imported graph forest has 997 parents, 36,568 first edges, 32 legacy structural continuation nodes, 256 depth-two edges, and 36,792 legacy leaves, with maximum depth two. K3P quartics terminate all 32 continuations at depth one, so the active minimal proof has 36,568 terminal rows. The depth-two replay remains as a redundant graph-and-algebra audit. There are zero missing/duplicate children, cycles, broken transports, or unresolved rows.

The 56 all-edge K3P quartics split as 24 formerly transported quartic rows plus the 32 legacy continuation nodes, all regenerated and checked in the active K3P coordinate system.

The producer-independent replay passed with canonical payload `91a29973bcd333c05270ffd379427a8c949285f791fd5db299a83c46934e82b1`. The mutation suite rejected all 20 mutations, with payload `f770cecb54b459b3ff11d85b0df780e10c07b8fff92dee553ce0feb91120a9c1`. The sealed manifest payload is `b35346386300bb3f0816a2bbe671280f92f28ee1c5fa2eb9829e6f9fe0863c39`.

16 Probe census

The 176 anchors comprise 143 isomorphisms and 33 triangle relations in 39 canonical classes.

category	one port	two ports
displayed quartet	27,758	511,266
K3P tree–sunlet SOS	99	576
isomorphism	1,915	30,969
triangle	192	1,760
raw rows	29,964	544,571
equality survivors	2,107	32,729
unresolved	0	0

The transport package has 67,741 exact transport rows and 4,379 parent restrictions, with zero incoherent choices. Ordered ledger roots and compressed-ledger SHA-256 values are in `probes/ONE_PORT_PROBE_MANIFEST.json`, `probes/TWO_PORT_PROBE_MANIFEST.json`, and `probes/GLOBAL_TRANSPORT_MANIFEST.json`.

The coherence layer is sealed against the complete replacement K3P restoration theorem by `probes/RESTORATION_MANIFEST.json`, schema `k3p-restoration-manifest-v2`, status `PASS`, with canonical payload `8c7eeeb77c09a64925fd604358b5660decca4699bd7a2fe04a0a52469a740fa7`. This binding distinguishes the 36,568 active K3P terminal rows from the 36,792 legacy/full-forest leaves and excludes historical K2P algebra.

17 Krawczyk certificate summary

The equality slice has 15 scaled pivot variables, 49 frozen direct parameters, and rational box radius 10^{-50}. The fifteen exact sparse equations have term counts

$$(5, 5, 4, 4, 3, 3, 7, 5, 4, 2, 5, 3, 7, 4, 3).$$

The maximum scaled residual at the rational center is approximately $3.02838161 \times 10^{-89}$. The exact center Jacobian determinant is approximately $-3.05693250 \times 10^{-2}$. With exact $Y = J_F(y_0)^{-1}$,

$$\max_j \frac{|K_j - y_{0,j}|}{10^{-50}} = 9.74099938 \times 10^{-41}, \qquad \|I - Y J_F(X)\|_\infty \le 8.07702308 \times 10^{-47}.$$

Both inequalities are rigorous rational interval bounds. The first is a strict Krawczyk inclusion; the second gives uniqueness inside the selected slice.

18 Sharpness rank witnesses

map	selected local columns	point determinant (decimal summary)	uniform Neumann bound
W	$(0, 1, 2, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 18)$	$1.30773104 \times 10^{-336}$	$1.54315210 \times 10^{-45}$
W'	$(0, 1, 2, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17)$	$-6.74132416 \times 10^{-325}$	$4.58271952 \times 10^{-45}$

The smallest certified edge-eigenvalue lower bounds are approximately 4.96448×10^{-10} and 1.39520×10^{-9}; the smallest continuous-time lower bounds are approximately 4.96448×10^{-10} and 1.39520×10^{-9}. Exact rational intervals for every edge, transition entry, inheritance probability, and minor are stored in the 9.9 MB Krawczyk certificate.

The exhaustive topology censuses are $W : (5, 2, 3)$, $W' : (7, 2, 5)$, and outer collision reference $(7, 0, 7)$.

19 Cherry determinant and all-n extension

Let ℓ be the old leaf replaced by the labelled cherry a, b, choose a retained outside leaf j, and write q' for the enlarged tensor. For $h = C, G, T$, the observable coordinates are

$$R_h = \frac{q'_{j=h,\, a=h,\, b=0,\, \text{others}=0}}{q'_{j=h,\, a=0,\, b=h,\, \text{others}=0}} = \frac{u_h q_{\ell=h,\, j=h,\, \text{others}=0}}{v_h q_{\ell=h,\, j=h,\, \text{others}=0}} = \frac{u_h}{v_h},$$
$$P_h = q'_{a=h,\, b=h,\, \text{others}=0} = u_h v_h q_{\ell=0,\, \text{others}=0} = u_h v_h.$$

The common factor in the ratio is strictly positive and $q_{\mathbf{0}} = 1$. Direct formal Laurent differentiation gives

$$\det \frac{\partial(R_C, P_C, R_G, P_G, R_T, P_T)}{\partial(u_C, v_C, u_G, v_G, u_T, v_T)} = \frac{8 u_C u_G u_T}{v_C v_G v_T}.$$

At $u = (2/5, 4/9, 3/7)$, $v = (3/7, 5/11, 4/9)$, this is $176/25$. The smallest exact strict physical margins at these two spectra are $83/630$ and $367/2772$. The positive inverse is $u_h = \sqrt{R_h P_h}$, $v_h = \sqrt{P_h / R_h}$. Each cherry adds exactly six dimensions, yielding $15 + 6(n - 3) = 6n - 3$.

20 Clean-room architecture and H21-01 repair

The historical clean-room verifier is preserved at SHA-256 `ee5e29a2cd795d9389e8e1257ebdb9eeaa4256fb5d03e0` `7f230bf82ba555ef91`.

It reproduces the H21-01 failure. The defect was mathematical, not a change in the expected orbit: it used rooted-DAG automorphisms instead of root-suppressed semi-directed mixed-graph automorphisms and did not conjugate the base target automorphism into the displayed frame. The corrected replay:

- binds five frozen inputs by hard-coded SHA-256 before parsing;
- reconstructs source and target incoming roles, repairs, physical ports, and port permutations from literal graphs;
- computes all seven H21 double cosets and the six nonisomorphic ones;
- replays all 38 raw Fourier-coordinate transports;
- binds each rank label to a square minor, selected rows and columns; and
- independently derives every target function-field rank upper bound.

The corrected verifier has zero Python assertion nodes and refuses optimized mode before reading inputs.

21 Mutation summary

The integrated primary verifier passes 28 of 28 active evidence entries. The integrated verifier for the main classification theorem also verifies the principal-domain and strict continuous-time equivalences in the article. Its fail-closed suite rejects 24 of 24 integrated mutations. The active component suites also include:

component	rejected	representative mutations
H21/fourteen-orbit gate	10	port/incoming changes, weakened rank bounds, nonsquare minor, selected-row change, optimized mode, input byte change
H21 adversarial re-audit (active wrapper command)	25	raw coverage, target conjugation, coordinate transport, graph-role and exact-identity corruptions
full four-port universe replay	6	coherent raw omission, isomorphism/triangle and restoration/quadratic reclassification, rank-upper forgery, quotient omission, optimized mode
graph-derived cut-topology replay	3	primitive-core change, coherently rebound mask/hash change, serial-topology change
cross-bridge local theorem	34	missing target, partition overlap, changed single/pair/cyclic identities, record-43/60 bindings
global cut transfer	30+32	circular dependency, crossing compatibility, side-blob physicality, target openness, tree-coloring counterexamples
integrated cut claim boundary	12	universal-pointwise substitution, claim-scope drift, circular dependencies, and ordinary/optimized modes
global analytic infrastructure	19	old uncapped bridge formula, selected-minor-only genericity, $C = T$, state permutation, missing domain inequality, rank-15 triangle, nonprimitive binomial, orientation-specific context
probe coherence	17	missing anchor/probe/parent, reversed order, broken transport, new/inconsistent triangle, $C = T$, optimized mode
probe all-row semantic replay	7	coherently resealed nonincidence transport and wrong marginal, quartet and six-circuit semantics, incomplete site profile, descriptive transport-scope drift, mixed Bernstein signs
sharpness	18	rooting census, graph status, Krawczyk data, interval rank, physical margins, cherry determinant and persistence
restoration	20	missing/duplicated child, wrong parent, K2P algebra, altered SOS/quadratic/quartic, broken marginal transport
integrated containment-classification verifier	24	universal-pointwise substitution, rank-15 triangle, proper directed containment, stale restoration binding, K2P leakage, reversed CT transfer, missing coherent boundary transport, omitted or reclassified full-universe rows, semantic probe omissions and survivors, weakened all-n sharpness, submission-ready and optimized-mode bypasses

Each campaign records the expected and observed diagnostic; a stale hash alone is not accepted as the intended mathematical rejection when the suite is designed to rebind mutated inputs.

22 Reproduction commands

From the project root, the component-level commands used by this draft are:

```
.venv/bin/python reproducibility/run_release_suite.py quick
.venv/bin/python reproducibility/run_release_suite.py full

bash reproducibility/verify_primary.sh
bash clean_room/verify_clean_room.sh
bash \
```

```
  cut_recovery/strong_crossbridge/topology_regeneration/verify_all.sh

.venv/bin/python \
  cut_recovery/strong_crossbridge/final_certificate/build_final_certificate.py
.venv/bin/python \
  cut_recovery/strong_crossbridge/final_certificate/verify_final_certificate.py
.venv/bin/python \
  cut_recovery/strong_crossbridge/global_transfer/verify_release.py

.venv/bin/python \
  three_port/literal_separator_v2/generate_literal_separator_v2.py
.venv/bin/python \
  three_port/literal_separator_v2/verify_literal_separator_v2.py
.venv/bin/python \
  three_port/literal_separator_v2/test_literal_separator_v2_mutations.py

.venv/bin/python restoration/regenerate_k3p_restoration.py --fresh
.venv/bin/python restoration/verify_k3p_restoration.py
.venv/bin/python restoration/test_k3p_restoration_mutations.py
.venv/bin/python restoration/test_restoration_report_portability.py

.venv/bin/python \
  four_port_atlas/full_universe_replay/generate_full_four_port_replay.py
.venv/bin/python \
  four_port_atlas/full_universe_replay/verify_full_four_port_replay.py
.venv/bin/python \
  four_port_atlas/full_universe_replay/test_full_four_port_mutations.py

.venv/bin/python probes/regenerate_k3p_probes.py
mkdir -p release/work/regeneration_ephemeral
.venv/bin/python probes/verify_k3p_probes.py --output \
  release/work/regeneration_ephemeral/K3P_PROBE_INDEPENDENT_VERIFICATION.json
.venv/bin/python probes/verify_k3p_probes_semantic.py
.venv/bin/python probes/test_k3p_probe_mutations.py --no-write-report
.venv/bin/python probes/seal_probe_manifests.py

.venv/bin/python global_infrastructure/generate_global_infrastructure.py
.venv/bin/python global_infrastructure/verify_global_infrastructure.py
.venv/bin/python global_infrastructure/test_global_infrastructure_mutations.py

.venv/bin/python sharpness/independent_krawczyk_replay.py
.venv/bin/python sharpness/independent_topology_alln_replay.py
```

The first two commands run the integrated verifier for the main classification theorem and its 24-mutation fail-closed suite. The longer producer commands are listed only for component replay. The canonical clean-checkout wrappers are `reproducibility/verify_quick.sh`, `reproducibility/verify_full.sh`, and the explicit-confirmation 54-command `reproducibility/verify_regenerate_all.sh`; the last includes the hour-scale probe producer and must be allowed to finish without a second invocation. No release-archive or submission gate is claimed here.

23 Artifact hashes

artifact	SHA-256
model_domain/primary_exact_evidence.json	ac21e4f795537f251377411841d670c1bad4ce06a69ed2 4f596252c11cb7afb6
three_port/primary_exact_evidence.json	152e7fa25d3c81fc7c27e46e93dbb6ee11e1c96be570d5 fe7a6b61eae2cc513d

artifact	SHA-256
three_port/literal_separator_v2/K3P_TREE_SUNLET_LITERAL_SEPARATOR_V2.json	5746324b4281a1738267ce116b3122e9156d48a6b5df37d5988451604e5f3557
bridge_fibre/K3P_BRIDGE_FIBRE_CERTIFICATE.json	bdd8e8f74a896b5031a50cad9ac97a82a03f4f821f2dc793cfb386415b1df532
marginals/K3P_MARGINAL_SUBMERSION_CERTIFICATE.json	9a90ba2e075fd20a4867e8a169aa933546b8fbca6feab23a2be7d25a24935d94
four_port_atlas/primary_exact_evidence.json	f4e02c74e4b4f9fbb063b13da65033a46ef45f2eecaad2b9efc1895b63afadab
clean_room/H21_01_TRANSPORT_AUDIT.json	b8a1c52a01ce0bb1b50a8ffaf78f5a300be52fde976671b2b1cc91e8fe2d7229
four_port_atlas/full_universe_replay/artifacts/FULL_FOUR_PORT_REPLAY.json	b07f57d1fc4e5ebd0b4bce2261931fa64399c791f539e02f46eff1146dcb76f4
four_port_atlas/full_universe_replay/INDEPENDENT_FULL_FOUR_PORT_VERIFICATION.json	543c38d9e74e4c9ee9f20e1ba2bd2de80e0476ddf0b544d7c7807025b9ed6fb8
four_port_atlas/full_universe_replay/FULL_FOUR_PORT_MUTATION_REPORT.json	4f360efc47f9213509092d40308fdc32e9e64bf8ee2a5d31ee6b912a546adaee
triangle_h14/K3P_H14_CONTEXT_CERTIFICATE.json	082f24cd26be08f248ddc6d0cf804059a27ad96477b06df0c56f8a89818bb9b9
cut_recovery/strong_crossbridge/topology_regeneration/CUT_TOPOLOGY_REGENERATION_REPORT.json	1e281437881cec0e1e44ed22be8376e1e22148d3bcb0a469afb872bb1515610f
cut_recovery/strong_crossbridge/final_certificate/STRONG_CROSSBRIDGE_FINAL_CERTIFICATE.json	643c29780219a538a5a127341ea91363aa5899d6f0f6fa1dac034889a7fdf06b
cut_recovery/strong_crossbridge/global_transfer/THEOREM_MANIFEST.json	eccd37323ee7b4c23959fc7905ed6c90f378c51e6b24108fa05058ae3f67de66
global_infrastructure/GLOBAL_INFRASTRUCTURE_MANIFEST.json	67e0dfa6052c6582a581931d909bb85386f0716721b63356d4e6d3c095dc82d9
global_infrastructure/MUTATION_CERTIFICATE.json	f55d6dfac9a8928fed3df279710f3e278a32e8815f10c5a25104cd1bd6cc8233
reproducibility/primary_gate_report.json	016789f7239d9c3787f64be9fb79ad5bab3236dd8830f3422de3687f4869c9f3
integrated classification gate report	89f77f1e95dbc006b0718b8fccc498b49c13679462f5208d3bc55b0ef9ea2c29
integrated classification mutation report	75220af80cd65f91b588cfa61e567bd1125690645457eb54503060b722721ef9
restoration/RESTORATION_MANIFEST.json	b7c53d68fc08f1bbde95802dd11ebcd12c52aef96edbb8f098b8f96caf8845c7
restoration/K3P_RESTORATION_THEOREM_REPORT.md	d20671c4d8b2c59e6159d9024e2bcc41c742898f8353eff531f1359fa8efc56e
restoration/K3P_RESTORATION_INDEPENDENT_VERIFICATION.json	7d826155d0e740e5456d5d8f9960b7c85a2f81adbab7d3370d69f3f2662622af
restoration/K3P_RESTORATION_MUTATION_CERTIFICATE.json	a2b4f111a52243863421b3d3090d62b899de6cf7ef110bee5e1a7a1e7bae7447
probes/K3P_PROBE_COHERENCE_CERTIFICATE.json	5c0534a0844de7362830f6ec31c1193db1b7f1c428224350b8d76d02f311b82f
probes/K3P_PROBE_SEMANTIC_VERIFICATION.json	4ecd5e5824385933085a2e103813a8bde58942f73ba988fb2f3113d7124eed2d
probes/K3P_PROBE_SEMANTIC_MUTATIONS.json	b4d4ed32cf1967928158df4b12606f379bb43d5798f971d54b4808282d8de016
probes/RESTORATION_MANIFEST.json	d6ca9f1234883d52d35b107f11d27c4fb6eda8382aa18664cf6af4f1fd7577e2
sharpness/K3P_SHARPNESS_KRAWCZYK_CERTIFICATE.json	8187174b3e0c0b3a0a55fa32595c211811c357dc223ada7a74b3033f7cae3941
sharpness/K3P_SHARPNESS_TOPOLOGY_ALL_N_CERTIFICATE.json	aa08837777445541398bed943881405530c9b5bb4f0451794a4b56b289beabc7

These are the frozen active hashes at this revision. The stored integrated report is the 59-binding artifact snapshot; the fresh fourteen-child replay is the separate executable gate recorded by the release runner. A later release manifest must supersede this typeset table if packaging changes any byte; no release archive or submission package is asserted here.